

Software Engineering V.IMP Questions

(EduShine Classes – Arman Ali)

Unit – 1 Introduction

- Q.1 Discuss the various characteristics of Good Software.
- Q.2 Explain the following statement “Software Engineering” is a layered Technology. (IMP)
- Q.3. Explain software characteristics in detail. Discuss the reasons of software crisis. (V.V.IMP)
- Q.4. Write short note on conventional engineering process.
- Q.5. Explain software quality attributes in detail. (V.V.IMP)
- Q.6. Explain software development life cycle. Discuss various activities during SDLC. (V.V.IMP)
- Q.7. Explain iterative waterfall and spiral model for software life cycle and discuss various activities in each phase. (V.IMP)
- Q.8. What is the need of SDLC? Discuss evolutionary development model in detail with the help of diagram.
- Q.9. List several software process paradigms. Explain how both waterfall model and prototyping model can be accommodated in the spiral process model.
- Q.10. Discuss the various McCall’s quality factors with quality triangle.

Unit – 2 SRS

- Q.1. Explain requirement engineering process. (V.V.IMP)
- Q.2. Explain requirement elicitation techniques in detail.

Q.3. Explain the feasibility studies. What are the outcomes? Does it have either implicit or explicit effects on software requirement collection?

Q.4. Discuss the importance of feasibility study. Also discuss its various types. (V.V.VIMP)

Q.5. Develop the level one DFD of library management system. (V.V.VIMP)

Q.6. Create a level-2 DFD of the Smart College Campus.(V.V.VIMP)

Q.7. Distinguish between a data flow diagram and a flow chart with example.

Q.8. Explain the decision table. Discuss the difference between decision table and decision tree.

Q.9. Write a detailed note on SEI CMM.

Q.10. Explain the SEI-CMM model. What do you mean by state fire fighting?

Unit – 3 Software Design

Q.1. Draw the software design framework and discuss the elements of design model.

Q.2. Illustrate the principles of software design. Discuss the characteristics of good software design. (V.V.VIMP)

Q.3. Explain data architectural and procedural design for a software.

Q.4. Explain about the structure chart and all its types with suitable example.

Q.5. Explain the term cohesion and coupling? Also explain the various forms of cohesion and coupling? (V.V.VIMP)

Q.6. What are the characteristics of a good design? Describe different types of coupling and cohesion. How design evaluation is performed? (V.V.IMP)

Q.7. What do you understand by token count? Explain Halstead software metrics in detail.

Q.8. Explain the need for software measures and describe various metrics.

Q.9. Explain software metric? Also explain the various metrics for the size estimation of a project.

Q.10. Draw the control flow graph and calculate the cyclomatic complexity with three methods and independent paths for: (V.V.IMP)

```
If A > 100
    THEN IF B > C
        THEN A = B
        ELSE A = C
    ENDIF
ENDIF
PRINT A
```

Unit – 4 Software Testing

Q.1. What is software testing? Discuss the objectives of it.(IMP)

Q.2. What is integration testing? What are various software integration techniques? (V.IMP)

Q.3. What is meant by system testing?

Q.4. What is regression testing? Discuss the process of test case prioritization in regression testing. (V.V.IMP)

Q.5. Distinguish among error, fault, and failure.

Q.6. Explain white box testing techniques.(V.IMP)

Q.7. Explain boundary value analysis and its significance with example.

Q.8. Write a short note on:

- Mutation testing
- Alpha and beta testing(IMP)
- Regression testing

Q.9. Explain code inspection, formal technical reviews (Peer reviews), and walk-through in detail.

Unit – 5 Software Maintenance

Q.1 Why is Software Maintenance required? Discuss with examples.

Q.2. Explain software re-engineering and the steps involved in the re-engineering process. (V.VIMP)

Q.3. What do you mean by the term software re-engineering? Why is it required?

Q.4. Discuss software configuration management and various tasks in SCM process. Explain version control and various types of project risks.(V.VIMP)

Q.5. Write briefly on CASE tools.

Q.6. Write briefly on:

- CASE
- Software complexity measure

Q.7. Write short notes on:

- Function Point
- COCOMO
- Defect, Fault, Failure

Q.8. Discuss COCOMO model in detail. Also explain the term person-month (PMO).(V.IMP)

Q.9. Write short note on software risk analysis and management.

Q.10. What is risk management? How are project risks different from technical risks?(V.IMP)

Solution Link - [Click here](#)

CLASSES

By- ARMAN ALI

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B. TECH.
(SEM IV) THEORY EXAMINATION 2018-19
SOFTWARE ENGINEERING

Time: 3 Hours**Total Marks: 70****Note: 1.** Attempt all Sections. If you require any missing data choose suitably.

SECTION A

1. Attempt all questions in brief.**2 x 7 = 14**

- (a) List the process maturity levels in SEI's CMM.
- (b) Compare evolutionary and throw away prototyping?
- (c) Draw the Context level DFD for the Safe home Software.
- (d) Distinguish between horizontal and vertical partitioning
- (e) Distinguish between verification and validation
- (f) Write short notes on equivalence partitioning
- (g) Define software re-engineering

SECTION B

2. Attempt any three of the following:**7 x 3 = 21**

- (a) Explain iterative waterfall and spiral model for software life cycle and discuss various activities in each phase.
- (b) Describe how software requirements are documented? State the importance of documentation.
- (c) Explain data architectural and procedural design for a software.
- (d) Describe decomposition levels of abstraction and modularity concepts in software design.
- (e) Define black box testing strategy. What do you mean by integration testing? Explain their outcomes.

SECTION C

3. Attempt any one part of the following:**7 x 1 = 7**

- (a) List several software process paradigms. Explain how both waterfall model and prototyping model can be accommodated in the spiral process model.
- (b) Which is more important-the product or process? Justify your answer

4. Attempt any one part of the following:**7 x 1 = 7**

- (a) Explain the feasibility studies. What are the outcomes? Does it have either implicit or explicit effects on software requirement collection
- (b) Narrate the importance of software specification of requirements. Explain a typical SRS structure and its parts.

5. Attempt any one part of the following:**7 x 1 = 7**

- (a) Explain about the various design concepts considered during design.
- (b) What are the characteristics of a good design? Describe different types of coupling and cohesion. How design evaluation is performed?

6. Attempt any one part of the following:**7 x 1 = 7**

- (a) What do you mean by boundary value analysis? Give two examples of boundary value testing.
- (b) What do you mean by system testing? Explain in detail

7. Attempt any one part of the following:**7 x 1 = 7**

- (a) Explain the need for software measures and describe various metrics
- (b) Write briefly on
 - i) CASE
 - ii) Software complexity measure.



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BTECH
(SEM VI) THEORY EXAMINATION 2021-22
SOFTWARE ENGINEERING

Time: 3 Hours**Total Marks: 100****Note:** Attempt all Sections. If you require any missing data, then choose suitably.**SECTION A****1. Attempt all questions in brief.****2*10 = 20**

Q.No.	Questions	CO
(a)	Enlist characteristics of software.	1
(b)	Define SDLC.	1
(c)	Define software crisis.	2
(d)	Compare ISO 9000 and SEI-CMM.	2
(e)	Examine methods of finding cyclomatic complexity with example.	3
(f)	Explain Function oriented Design.	3
(g)	List the points of differences between Verification and Validation.	4
(h)	What are stub and driver?	4
(i)	Differentiate between adaptive and corrective maintenance?	5
(j)	List the points of differences between software Re-engineering and Reverse engineering.	5

SECTION B**2. Attempt any three of the following:****10*3 = 30**

Q.No.	Questions	CO
(a)	Explain prototyping model of SDLC. What are its advantages over conventional model? Explain with diagram.	1
(b)	Create a level-2 DFD of the Smart College Campus.	2
(c)	Differentiate between the features of Top-down and Bottom-up approaches of software design along with its advantages and disadvantages.	3
(d)	What is regression testing? Discuss the process of test case prioritization in regression testing?	4
(e)	Why is Software maintenance required? Explain types of maintenances with examples.	5

SECTION C**3. Attempt any one part of the following:****10*1 = 10**

Q.No.	Questions	CO
(a)	Discuss Spiral Model for Software development life cycle and highlight the Risk analysis in this context.	1
(b)	What is the need of SDLC? Discuss evolutionary development model in detail with the help of diagram.	1

4. Attempt any one part of the following:**10 *1 = 10**

Q.No.	Questions	CO
(a)	Discuss about decision tables and its components. Create a decision table. for the following scenario; a bookstore gets a trade discount of 25% for order more than 6 books;for order from libraries and individuals, 5% allowed on orders of 6-19 copies per book title; 10% on orders for 20-49 copies per book title; 15% on orders for 50 copies or more per book title.	2



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BTECH
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SOFTWARE ENGINEERING

(b)	Discuss the importance of software specification Document. And also explain the typical IEEE format of SRS document.	2
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5. Attempt any one part of the following: 10*1 = 10

Q.No.	Questions	CO
(a)	With the help of example illustrate the concept of modularity and discuss why is “Low coupling –High cohesion is better for good software” along with the concept of functional independence.	3
(b)	What are the various software design strategies? Analyze the points of difference between Function Oriented Design and Object-Oriented Design.	3

6. Attempt any one part of the following: 10*1 = 10

Q.No.	Questions	CO
(a)	Explain Halstead software metrics in detail and mention what do you understand by token count?	4
(b)	Discuss the differences between black box and white box testing and explain how these techniques can be used simultaneously to test a system.	4

7. Attempt any one part of the following: 10*1 = 10

Q.No.	Questions	CO
(a)	Explain Risk management in detail. Also discuss the points that differentiate project risk from technical risk.	5
(b)	What is cost analysis in context of software? Explain COCOMO model with the help of schematic diagram.	5

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B. TECH.
(SEM VI) THEORY EXAMINATION 2022-23
SOFTWARE ENGINEERING

Time: 3 Hours

Total Marks: 100

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

2 x 10 = 20

- (a) Define generic software with example.
- (b) Define software components.
- (c) Mention any two non-functional requirements on software to be developed.
- (d) What is meant by software prototyping?
- (e) What do you mean by horizontal and vertical partitioning?
- (f) What is the software architecture?
- (g) Distinguish between verification and validation.
- (h) Distinguish between alpha and beta testing?
- (i) Mention the advantages of CASE tools.
- (j) Define adaptive maintenance.

SECTION B

2. Attempt any three of the following:

10x3=30

- (a) Iterative waterfall and spiral model for software lifecycle clean and discuss various activities in each phase.
- (b) What are the problems faced in software engineering? Explain each in detail.
- (c) Explain about the structure chart and all its types with suitable example.
- (d) Explain boundary value analysis and its significance with example.
- (e) Discuss Software Configuration Management and various tasks in SCM process. Explain version control and various types of project risks.

SECTION C

3. Attempt any one part of the following:

10x1=10

- (a) Explain software characteristics in detail. Discuss the reasons of software crisis.
- (b) Discuss the various Mc Call's quality factors with quality triangle.

4. Attempt any one part of the following:

10x1=10

- (a) Explain the SEI-CMM model. What do you mean by state of fire fighting.
- (b) What are the various stages of requirement engineering process? Explain it with diagrammatic representation.

5. Attempt any *one* part of the following: 10x1=10

- (a) Draw the software design framework and discuss the elements of design model.
- (b) Illustrate the principles of software design. Discuss the characteristics of good software design.

6. Attempt any *one* part of the following: 10x1=10

- (a) Draw the control flow graph and Calculate the cyclomatic complexity with three methods and independent paths for the given code-

```
IF A = 100  
THEN IF B > C  
THEN A = B  
ELSE A = C  
ENDIF  
ENDIF  
PRINT A
```

- (b) Discuss the various types of structural testing techniques with example of each.

7. Attempt any *one* part of the following: 10x1=10

- (a) Draw the general model of software re-engineering and elaborate it.
- (b) Write short notes on:
 - (i) Function Point
 - (ii) COCOMO
 - (iii) Defect, Fault, Failure



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BTECH
(SEM VI) THEORY EXAMINATION 2023-24
SOFTWARE ENGINEERING

TIME: 3 HRS**M.MARKS: 100****Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief.**

a.	Define the term Software Engineering.	02
b.	Discuss the various characteristics of a software.	02
c.	Explain the need of an SRS.	02
d.	Explain stubs and drivers.	02
e.	Discuss LOC? List two advantages and disadvantages of LOC.	02
f.	What is Pseudo Code? How it differs from Algorithm?	02
g.	Describe the importance of white box testing.	02
h.	Explain Error, Fault and Failure.	02
i.	List any two reasons for increase in the software costs.	02
j.	Discuss the need of Risk Management in software engineering.	02

SECTION B**2. Attempt any three of the following:**

a.	Illustrate the statement "Software engineering is layered technology".	10
b.	Discuss the importance of Feasibility Study. Also discuss its various types.	10
c.	Explain Code Inspection, Formal Technical Reviews (Peer Reviews) and Walk Through in detail.	10
d.	Write a short note on: Mutation testing Alpha & Beta testing Regression testing	10
e.	What do you mean by the term software re-engineering? Why is it required?	10

SECTION C**3. Attempt any one part of the following:**

a.	Explain Software Quality Attributes in detail	10
b.	Explain SDLC. Also discuss various activities during SDLC	10

4. Attempt any one part of the following:

a.	Explain Requirement Elicitation techniques in detail.	10
b.	Compare SEI CMM Model and ISO 9000 Model. Also discuss five levels of CMM.	10

5. Attempt any one part of the following:

a.	Explain the term Cohesion and Coupling? Also explain the various forms of cohesion and coupling?	10
b.	Explain software metric? Also explain the various metrics for the size estimation of a project.	10

6. Attempt any one part of the following:

a.	A program reads an integer number within the range [1,100] and determines whether it is a prime number or not. Design test cases for this program using BVC, robust testing, and worst-case testing methods.	10
b.	What is Integration Testing? Explain different approaches used for integration testing.	10

7. Attempt any one part of the following:

a.	Discuss the need of maintenance. Also discuss various categories of maintenance.	10
b.	Discuss COCOMO model in detail. Also explain the term Person Month (PM).	10